

Illustration 5-52: Helium Gas Hose Connected to Nupro Valve

**NOTICE**

Possible quench may occur if the gas helium flow rate of 140 cfh is exceeded.

6. Using the scale marked 20 to 140 for helium, set the flow rate on the flow meter to 80 cfh. (It may take several attempts to clear all the ice from the Sav-Con Connector.)
7. Open Helium Vent Valve V2 so as not to increase pressure in magnet.
8. Slowly open the Nupro Valve, and watch the pressure on the magnet to ensure it stays below 0.5 psi.
9. Let the helium gas flow over the Sav-Con connector for one (1) minute.
10. Turn off the gas helium and see if the shim lead will lower into the Sav-Con. Make sure that the Shim Lead is aligned with the keyway.

**NOTICE**

Possible quench may occur if the gas helium flow rate of 140 cfh is exceeded.

11. If the Shim Lead will not seat, continue with the helium gas and set the flow meter to 140 cfh for two (2) minutes. Repeat if necessary.

12. Once a good fit is achieved on the Sav-Con Connector, verify ample continuity exists on all the correction coils, then disconnect the helium gas line from the Shim Lead ensuring the Vent Valve V2 and Nupro Valve are completely closed. (See [Illustration 5-50](#).)
13. If unable to clear the Save-Con Connector, contact the Magnet Plant in Florence, South Carolina for further instructions.

9.2.4.2 Shim Lead Connector and Vertical Penetration Deicing - Level Two



NOTICE

Make sure the Vertical Penetration is clear of any obstructions before inserting/extracting any tools/apparatus to prevent possible magnet damage. When removing ice in the Vertical Penetration, avoid contacting the Instrumentation Lead Connector with ice clearing tools. Contact could damage the lead or connector.

1. Either turn the Magnet Monitor power off, or disable the Magnet Monitor's Master Alarm and Vessel Pressure Control. (This will prevent false alarms and long-term heater operation.)
2. Make sure the Shim Lead is disengaged, then engage the Shim Lead in conformance with Shim Lead Engagement and Disengagement.
3. Perform Shim Coil resistance checks per [Chapter 4, Magnet Electrical Checks](#).
 - If the checks are successful, deicing is not required.
 - If the checks are **not** successful or the Shim Lead cannot be engaged, proceed to the next step.
4. Ramp down the magnet if at field using the 50 ft. (15.24 m) Auxiliary Rampdown Cable (2217315) in conformance with [Magnet Rampdown to Zero Field](#).
5. Turn off the Cryocooler to prevent ice migration into the Recondenser.
 - a. Turn the Coldhead Drive Switch On (up position), located on the Compressor's back panel.

Illustration 5-53: Views of CSW-71D Compressor

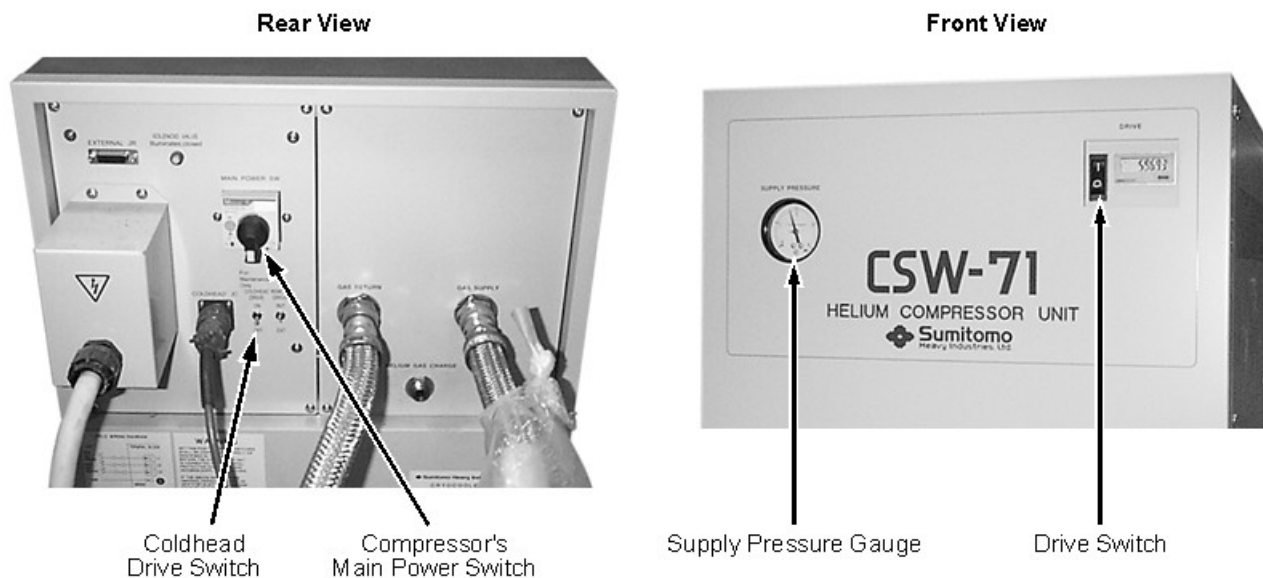
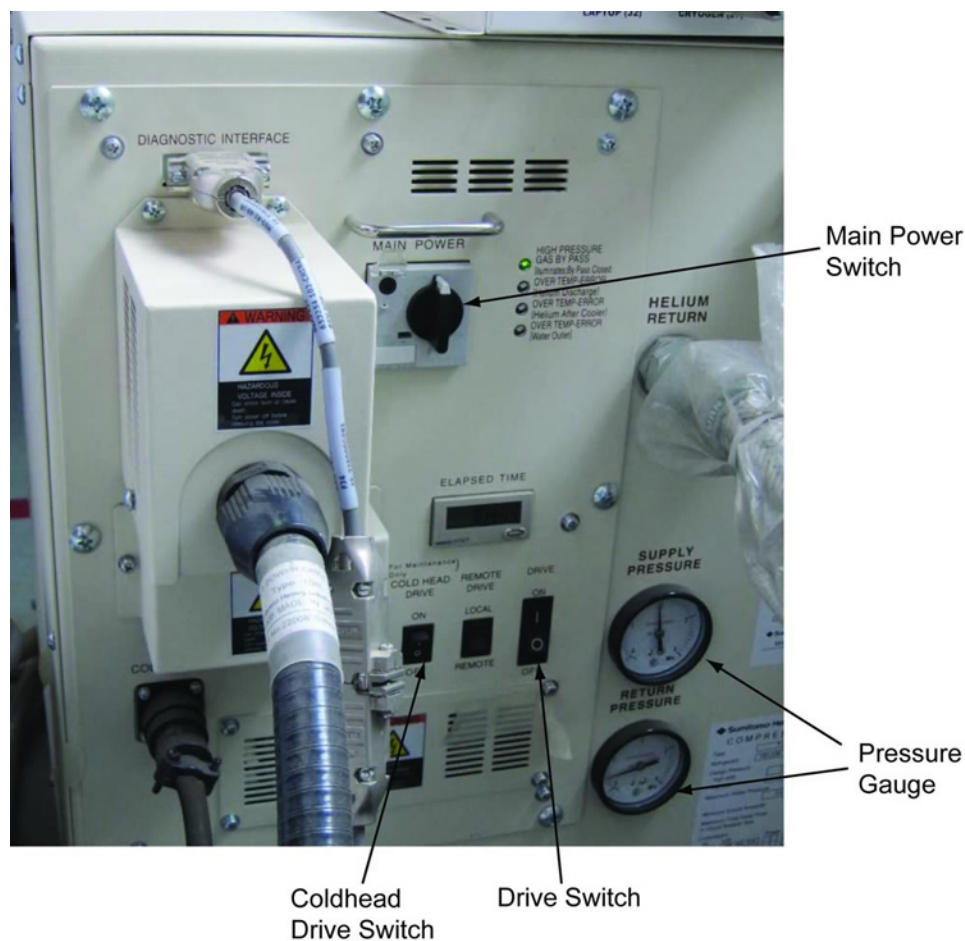


Illustration 5-54: View of the F50 Compressor



- b. Place the Drive Switch on the Compressor's front panel in the O (Off) position.

- c. Run the Coldhead for only 10 seconds with the Compressor turned off to prevent pressure build-up in the Coldhead, then turn OFF the Coldhead Drive Switch and the Compressor's Main Power Switch.

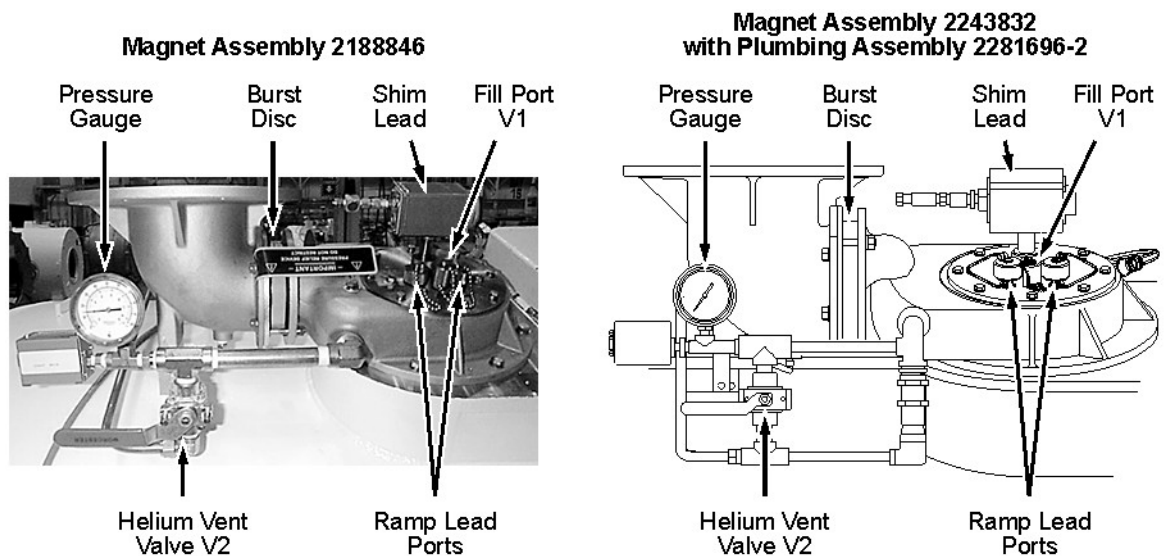


WARNING

ELECTRICAL SHOCK HAZARD!
CONTACT WITH CONNECTORS LEADING TO AN ENERGIZED COMPRESSOR CAN CAUSE ELECTRICAL SHOCK.
WHENEVER THE INPUT POWER TO THE COMPRESSOR IS DISCONNECTED, FOLLOW LOTO PROCEDURES TO MAKE SURE POWER IS NOT SUPPLIED TO THE COMPRESSOR.

- d. Disconnect and LOTO input power to the Compressor. Verify that no voltage is present by using a DVM or equivalent measuring device.
6. Vent the magnet down to 0.2 psig by opening/closing Vent Valve V2.

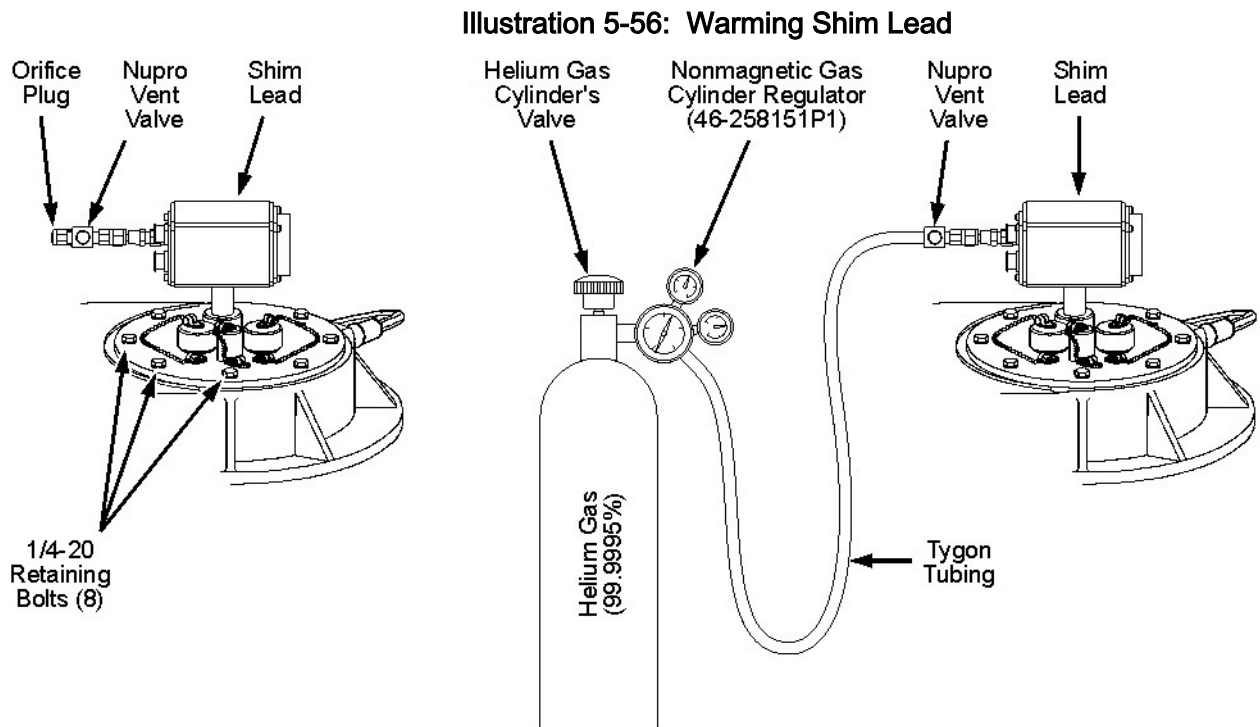
Illustration 5-55: Helium Vent Plumbing



Note: Stainless Steel Plumbing Assembly 5175534 is identical to Plumbing Assembly 2281696-2 for the purposes of this illustration.

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7. Assemble the regulator and Tygon tubing to a helium cylinder.
 - a. Remove the orifice plug and nut at the end of the Shim Lead Nupro Vent Valve.
 - b. Assemble the other end of the Tygon tube over the exhaust stem.



8. Partially open Vent Valve V2, and immediately open the Nupro Vent Valve. Blow warm helium gas through the Shim Lead at 4 psig for five (5) minutes with the Shim Lead engaged/partially engaged.
9. Attempt to engage the Shim Lead Connector. If successful:
 - a. Close the Nupro Vent Valve and Vent Valve V2
 - b. Perform Shim Coil resistance checks per [Chapter 4, Magnet Electrical Checks](#).
 - c. Remove the Tygon tubing, and replace the orifice plug and nut.
 - If the checks are successful, deicing is not required.
 - If the checks are **not** successful or the Shim Lead **cannot** be engaged, proceed to the next step.
10. Disengage the Shim Lead, and lock it in the **Up** position in conformance with Shim Lead Engagement and Disengagement.
11. Make sure magnet pressure is 0.2 psig, venting the magnet as required.

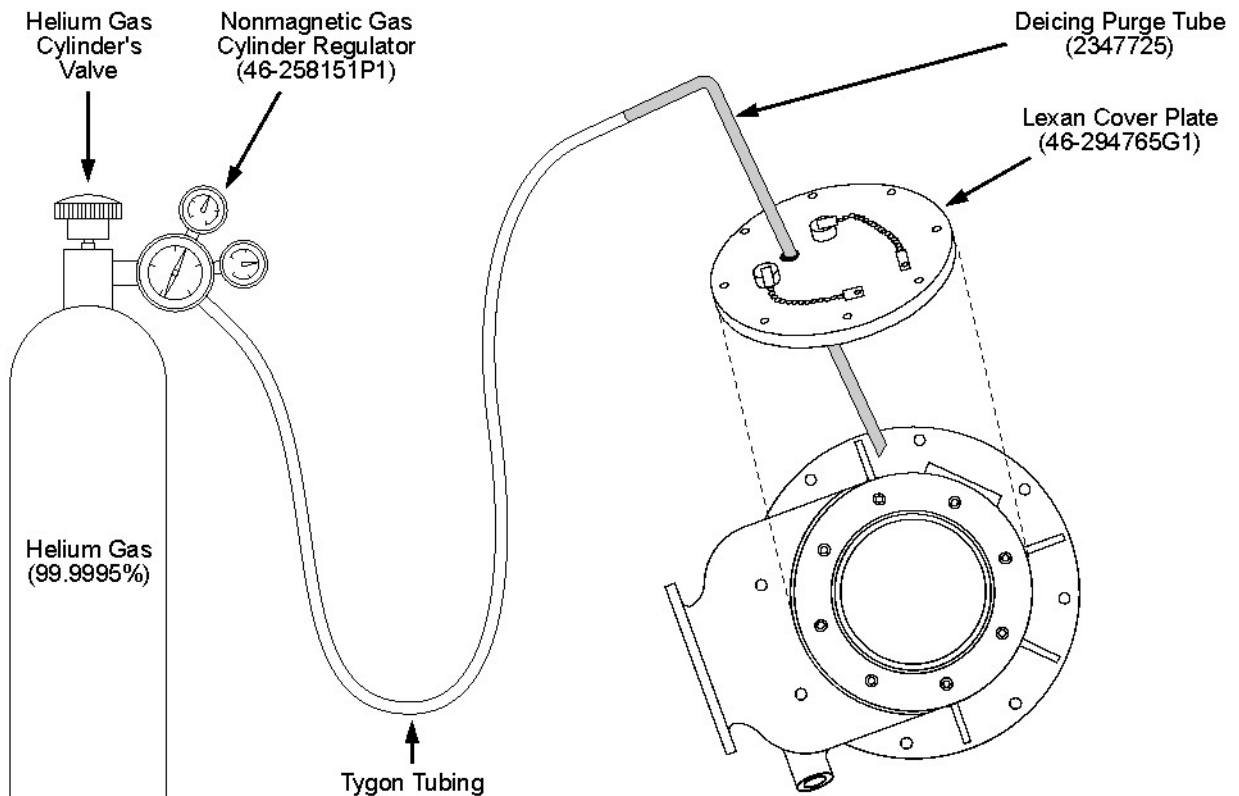


NOTICE

Perform the next two steps rapidly to prevent cryo-pumping of air into the Vertical Stack. Make sure Lexan Cover Plate 46-294765G1 is on the Service Platform and the holes are plugged before removing the Shim Lead Assembly.

12. Loosen and remove the eight 1/4-20 retaining bolts (identified in [Illustration 5-56](#)), and remove the Shim Lead Assembly.
13. Immediately cover the Vertical Penetration with the Lexan Penetration Cover Plate 46-294765G1, and align the scribe mark toward the Vent Adapter.
14. Secure the Lexan Penetration Cover Plate onto the Vertical Penetration with the eight 1/4-20 retaining bolts removed earlier.
15. Attach Deicing Purge Tube 2347725 to the end of the Tygon tubing, and purge the tube at 4 psig.
16. Remove the plugs on the Cover Plate, and insert the Deicing Purge Tube through the hole closest to the Shim Lead Connector.

Illustration 5-57: Purge Tube Insertion



17. Remove ice by directing an 8 to 10 psig flow of warm helium gas onto the Shim Lead connector inside the Vertical Penetration and other affected areas while Vent Valve V2 is partially open.
18. Shine the flashlight through the Lexan Penetration Cover Plate, and verify that all ice is removed inside.
19. Remove any remaining ice on the inside of the Vertical Penetration. (Some bending of the Deicing Purge Tube may be required.)

20. Shine the flashlight through the Lexan Penetration Cover Plate again. When all ice is removed, close V2, and remove the Lexan Cover Plate.
21. Temporarily disconnect the Instrumentation Lead Connector using the Connector Insertion/Removal Tool (46-281934P1).
22. Direct an 8 to 10 psig flow of warm helium gas at the connector until all ice is removed, then reconnect the Instrumentation Lead Connector.
23. Replace the Shim Lead Assembly in conformance with [Shim Lead and Baffle Replacement](#).
24. Perform Shim Coil resistance checks per [Chapter 4, Magnet Electrical Checks](#). If the checks are not successful, replace the Shim Lead Sav-Con Connector in conformance with [Sav-Con Connector Replacement](#).



NOTICE

Perform all MRU checks in upon completion of Vertical Penetration ice removal to make sure no Instrumentation Lead Connector damage has occurred. Correct any damage found by reconnecting the Instrumentation Lead Connector or replacing the Instrumentation Lead Assembly.

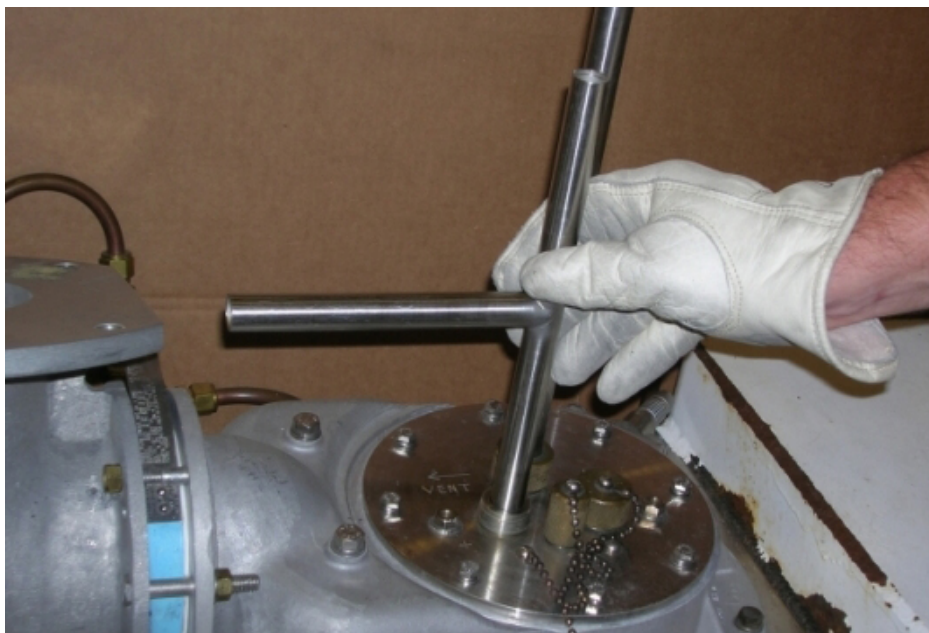
25. Check the Instrumentation Lead Connector by performing Magnet Rundown Unit (MRU) Checks.
26. After completing the above step, remove LOTO, connect input power to the Compressor, and turn on the Cryocooler.
27. Leak check the magnet in conformance with [Leak check the magnet in conformance with Chapter 4, Temperature, Pressure and Leak Troubleshooting](#). Repair any leaks found.
28. Return the Magnet Monitor to normal operation by re-enabling the Magnet Monitor's Master Alarm and Vessel Pressure Control or by turning Magnet Monitor power on.

9.2.4.3 Ramp/Fill Port Deicing

This procedure is performed when the quantity of ice in the magnet prevents interaction with the ramp pins or the fill port in the penetration. Ice build-up is usually due to an air leak somewhere. It is advisable to locate the leak and correct it.

1. Reduce the vessel pressure down to ≤ 0.2 psi by opening Vent Valve V2.
2. Slowly insert the Noninsulated Coring Tool (2280035) into either the (+) ramp port or (-) ramp port.
 - a. Once the ice buildup stops the tool, rotate the tool back and forth while gently pushing down. (The sawing action will cut through the ice down to the top of the ramp pin.)

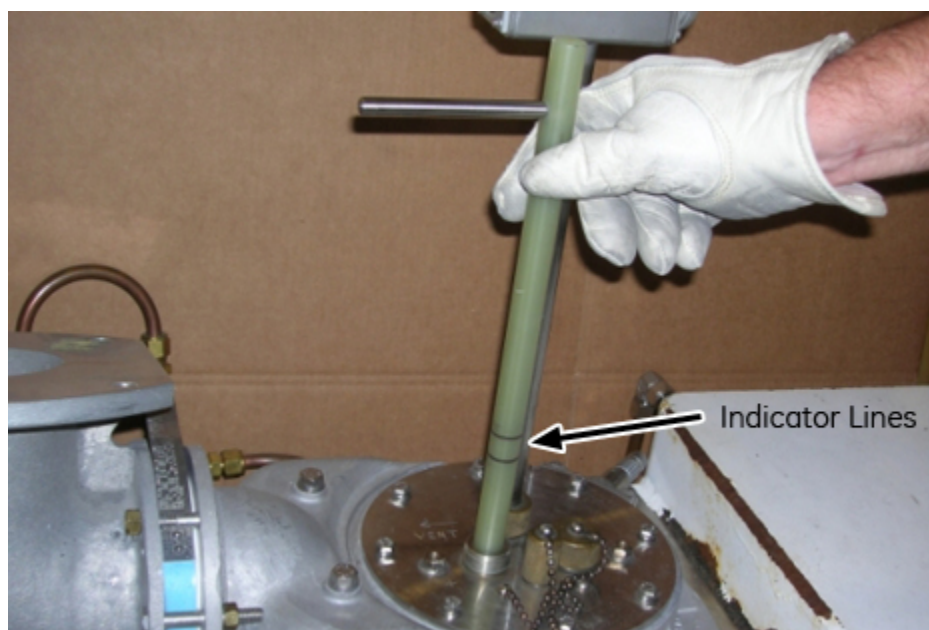
Illustration 5-58: Ramp Port Deicing - Noninsulated Tool



- b. Remove the tool once it has hit bottom, and use Insulated Ramping Port Tool (5248857) with the same motion. (The tool will cut through the remaining ice to the bottom of the pin.)

NOTE: There is an indicator line on the tool that allows the user to determine how deep to go in the penetration. There are two lines, one for LCC and one for SIV, clearly marked on the rod to avoid confusion. Do not allow the tool to go any deeper than this line.

Illustration 5-59: Ramp Port Deicing - Insulated Tool



3. Repeat the process for the other port, and remove the tool.
4. To deice the fill port, use the Noninsulated Tool (5248765) by inserting it slowly into the fill port of the magnet.
 - a. Once the ice build-up stops the tool, rotate the tool back and forth while gently pushing down. (The sawing action will cut through the ice down to the top of the fill tube.)

Illustration 5-60: Fill Port Deicing - Noninsulated Tool



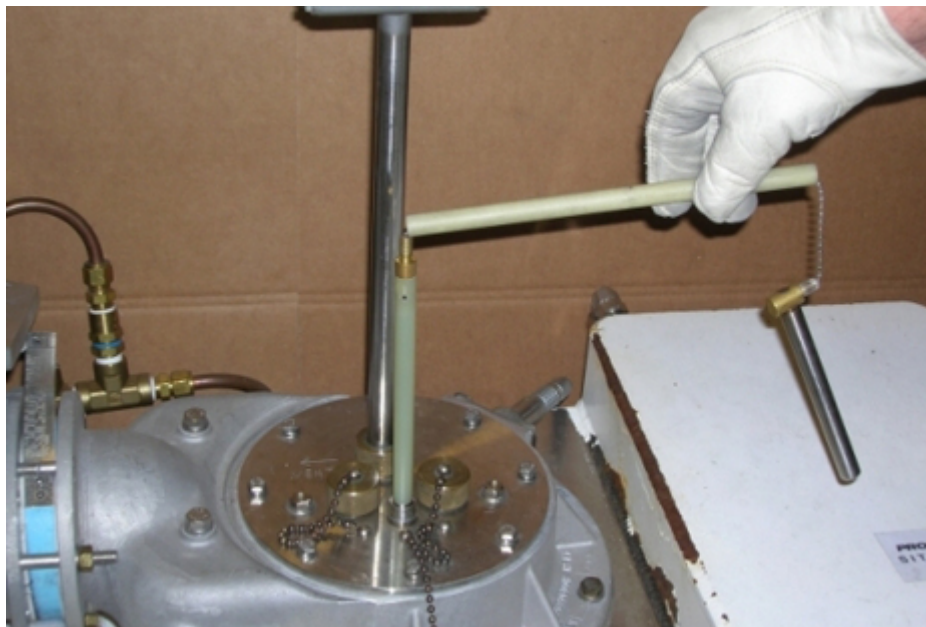
- b. Remove the tool once it has hit bottom, and use Insulated Fill Port Coring Tool (5248857) with the same motion. (The tool will cut through the remaining ice to the bottom of the fill tube.)

Illustration 5-61: Fill Port Deicing - Insulated Tool



- c. Remove the tool.
5. In the situation where a magnet is sited in a low ceiling environment, the Ramp Port Coring Tool (5248766) and the Fill Port Coring Tool (5248851) are used. The tool requires a three-step insertion process. The pictures below are for the fill port; the ramp port process is the same.
 - a. Insert the bottom piece.

Illustration 5-62: Fill Port Deicing in Low Ceiling Condition - Inserting Bottom Piece of Tool



- b. Thread the middle shaft onto the bottom shaft.

Illustration 5-63: Fill Port Deicing in Low Ceiling Condition - Inserting Middle Piece of Tool



- c. Screw the top onto the middle shaft.